

DATA ANALYTICS PROJECT 2

(EXPLORATORY DATA ANALYSIS)

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1. Introduction

This project centres on conducting Exploratory Data Analysis (EDA) to uncover patterns, trends, and data distributions within the dataset. The analysis involves computing descriptive statistics, detecting trends and outliers, and highlighting key insights obtained from the data.

2. Dataset Overview

The dataset contains information about customer orders, including product types, quantities, unit prices, total prices, payment methods, and customer sources.

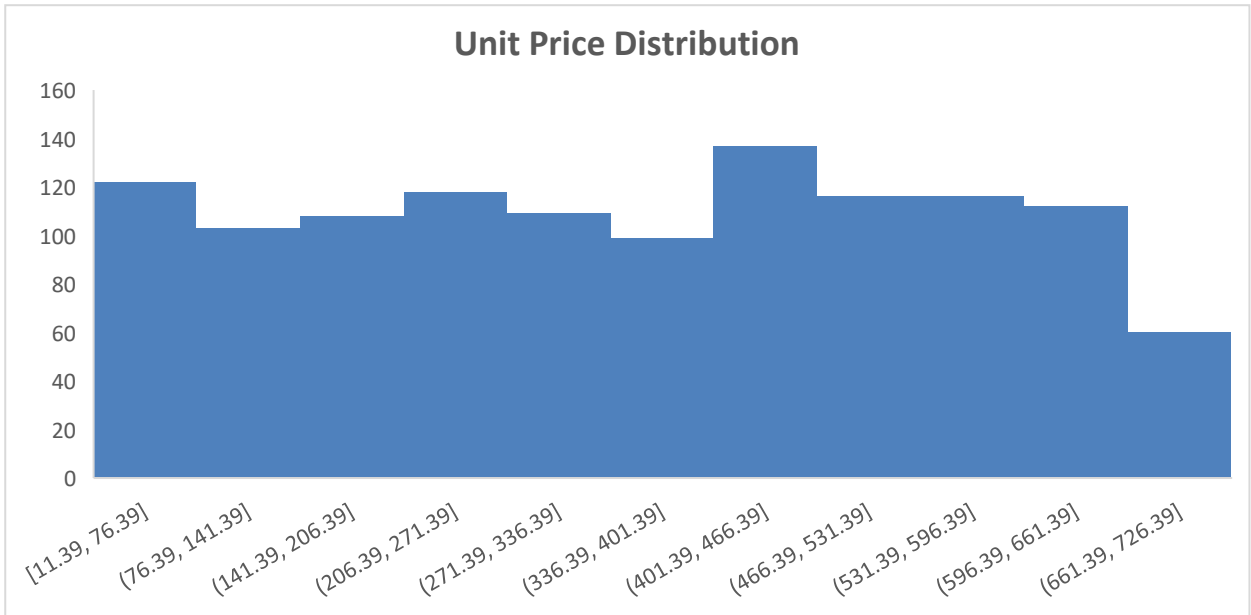
3. Descriptive Statistics

Metric	Quantity	Unit Price	Total Price
Mean	2.95	356.41	1,053.97
Min	1.00	11.39	11.39
Max	5.00	699.93	3,456.40
Count	1,200.00	1,200.00	1,200.00

The descriptive statistics indicate that customer orders are relatively consistent in quantity, averaging approximately 3 items per order. However, both unit and total prices exhibit significant variability, ranging from 11.39 to 699.93 and 11.39 to 3,456.40 respectively.

This wide spread suggests a diverse pricing structure and heterogeneous order sizes, with outliers contributing to the overall dispersion of the dataset.

4. Unit Price Distribution



The histogram shows a wide dispersion of unit prices, indicating high variability across products. Most values are concentrated in the mid-range (200–600), suggesting that the majority of items are moderately priced.

The relatively even spread of frequencies across bins reflects a broad, non-centralized distribution, while the presence of values at both lower (<100) and upper (>650) ranges indicates tail effects and outliers.

Overall, the pattern suggests a diversified, multi-tier pricing structure with moderate central concentration and extended extremes.

5. Trends & Patterns

The analysis reveals clear relationships between pricing, customer behaviour, and purchasing patterns:

- **Price vs Quantity:** Lower-priced products are generally purchased in higher quantities, while higher-priced products are purchased in smaller quantities. This indicates that customer demand is strongly influenced by pricing.
- **Product Demand:** Frequently purchased products such as chairs, desks, and tablets indicate higher demand, while expensive items like laptops tend to have lower purchase frequency.
- **Customer Channels:** A large proportion of orders originate from social media platforms such as Instagram and Facebook, suggesting that these platforms are key drivers of customer engagement.
- **Coupon Usage:** Coupons such as *FREESHIP* and *WINTER15* are widely used, indicating that discounts and promotional offers significantly influence purchasing behaviour.

Overall, these patterns indicate a price-sensitive demand structure, where affordability, promotions, and digital engagement significantly influence purchasing behaviour.

6. Outliers

The dataset contains outliers at both ends of the distribution, including extremely low prices (11.39) and high values above 600. These extreme observations contribute to overall variability and may represent exceptional transactions or data anomalies.

7. Key Observations

- Customers buy around 3 items per order
- Price affects buying behaviour
- Discounts increase sales
- Some unusual values exist in the data.

8. Conclusion

The analysis demonstrates that customer purchasing behavior is largely influenced by pricing and promotional incentives. Lower-priced products drive higher volume, while higher-priced items contribute to value. Social media channels play a significant role in customer acquisition, and the presence of outliers highlights the importance of data validation. These insights can inform pricing and marketing strategies.